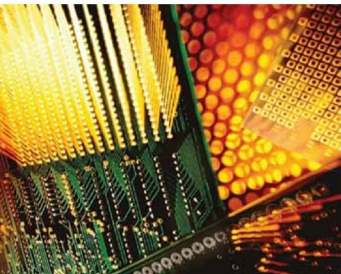


ATOMTRONICS TRYING TO REPLACE ELECTRONICS

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■ DR S.S. VERMA

Electronics is the field of electron movement in the circuits governed by the use of wires, silicon and electricity. All modern electronic devices contain transistors as the fundamental building blocks. Until recently, electronics had been based on a single property of electrons—their charge. But now physicists have begun to exploit another property—electron spin.

So-called spintronics promises to revolutionise electronics because it allows information to be encoded in an entirely new way. Though electronics is bestowed with a large number of inherited advantages, but in the era of quantum electronics it is facing new challenges. Because electrons lose any possible initial quantum state as they

bounce around through the energy-dissipating semiconductor or metallic systems, they are ill-equipped for quantum computing.

Atomtronics

We all have heard of electronics, mechatronics and maybe even spintronics, but the latest buzzword doing the rounds of physics labs is atomtronics—the science of creating circuits, devices and materials using ultra-cold atoms instead of electrons. What if atoms could be

used to perform the functions that are currently the province of electronic devices? The goal of atomtronics is to do just that by creating analogues to the common items found in electronic and spintronic devices.

Physicists Satyendra Nath Bose and Albert Einstein proposed in 1924 that large numbers of atoms could be chilled to the point that they joined together in a single quantum state, bringing subatomic effects to a scale accessible by laboratory experiments. But it wasn't until 1995 that scientists made a Bose-Einstein condensate (ultra-cold gas) using lasers to carefully cool rubidium-87 atoms down to temperatures less than a millionth of a degree above absolute zero. The 2001 Nobel Prize in Physics celebrated this accomplishment, which was also achieved using sodium atoms.

Atomtronics is a young and mostly theoretical field based on the idea that atoms in unusual quantum states of matter may provide an alternative to the tried-and-true electron for making useful devices. The field's proponents have drawn up blueprints for atomic versions of many traditional electronic components—from wires and batteries to transistors and diodes. The idea is to manipulate neutral atoms using lasers in a way that mimics the behaviour of electrons in wires, transistors and logic gates.

In atomtronics, the current carriers in electronics (electrons) are replaced with neutral, ultra-cold atoms; the semiconductor material that the electrons traverse is replaced with an optical lattice; and the electric potential difference, which induces the flow of electrons around the circuit, is replaced by a chemical potential difference.

Over the last decade or two, physicists have become masters at creating optical lattices in which atoms can be pushed, pulled and prodded at will. This optical property of atoms has not attracted much attention of workers but now people have begun a program to put tame atoms to work. The problem is that atoms don't behave like electrons. So, building the atomtronic equivalent of something even as straightforward as a simple circuit consisting of a battery and resistor in series requires some thinking out of the box.

The dynamics of atoms in optical lattices, which are basically crystals of light, has been studied theoretically and experimentally for many years now. It is just a further addition to this